

MH1101 Tutorial 4 (Week 5)

Reference: Sections 2.3, 3.1 (Lecture Notes)

1. Use the method of cylindrical shells to find the volume generated by rotating the region bounded by the given curves about the given axis.

(i) $y = x^2$, $y = 6x - 2x^2$; about $x = 0$.

(ii) $y = x^3$, $y = 8$, $x = 0$; about $x = 3$.

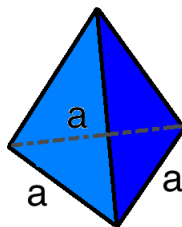
(iii) $x = 2y^2$, $x = y^2 + 1$; about $y = -2$.

2. Each integral represents the volume of a solid. Describe the solid.

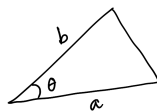
(i) $\int_0^3 2\pi x^5 dx$.

(ii) $\int_0^{\pi/2} 2\pi(x+1)(2x - \sin x) dx$.

3. Find the volume of a pyramid with height h and base an equilateral triangle with side a .



You can use the following sine formula for area of a triangle: Suppose θ is the angle between two sides (of length a and b respectively) of a triangle:



Then the area of the triangle is $\frac{1}{2}ab \sin \theta$. You can also use Heron's formula for the area of triangle.

4. Evaluate the integral using integration by parts.

(i) $\int x \cos 5x \, dx$

(ii) $\int \ln \sqrt{x} \, dx$

(iii) $\int_0^1 (x^2 + 1)e^{-x} \, dx$

(iv) $\int_1^{\sqrt{3}} \tan^{-1}(1/x) \, dx$

(v) $\int_0^{1/2} \cos^{-1} x \, dx$

5.

(a) Prove the reduction formula

$$\int \cos^n x \, dx = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int \cos^{n-2} x \, dx.$$

(b) Use part (a) to evaluate $\int \cos^2 x \, dx$.

Answer Keys. 1. (i) 8π (ii) $\frac{264}{5}\pi$ (iii) $\frac{16}{3}\pi$ 2. (i) the solid obtained by rotating the region bounded by the curves $y = x^4$ and $y = 0$, $0 \leq x \leq 3$, about the y -axis. (ii) the solid obtained by rotating the region bounded by the curves $y = 2x$ and $y = \sin x$, $0 \leq x \leq \pi/2$, about the line $x = -1$. 3. $\frac{\sqrt{3}}{12}a^2h$ 4. (i) $\frac{x}{5} \sin 5x + \frac{1}{25} \cos 5x + C$ (ii) $x \ln \sqrt{x} - \frac{x}{2} + C$ (iii) $-6e^{-1} + 3$ (iv) $\frac{\sqrt{3}\pi}{6} - \frac{\pi}{4} + \frac{\ln 2}{2}$ (v) $\frac{1}{6}(\pi + 6 - 3\sqrt{3})$ 5. (b) $\frac{1}{2} \cos x \sin x + \frac{x}{2} + C$